

LINUX LAB-Day1

User, Account & Group Management — Parts A–C

PART A — User and System Investigation (10 Minutes)

1. Determine the username of the account currently being used.

```
whoami
```

→ Shows the current username.

2. Display UID, primary GID and supplementary groups.

```
id
```

→ Shows the UID, primary GID and supplementary groups.

3. Display all users currently logged in.

```
who
```

→ Lists users currently logged in.

4. Determine what activities logged-in users are performing.

```
w OR ps -ef
```

→ w → Shows logged-in users and their current activity.

ps -ef → Lists running processes for all users.

```
gopika@kali:~$ whoami
gopika

gopika@kali:~$ id
uid=1000(gopika) gid=1000(gopika) groups=1000(gopika),4(adm),20(dialout),24(cdrom),25(floppy),27(sudo),29(audio),30(dip),44(video),46(plugdev),100(users),101(netdev),102(scanner),114(wireshark),119(kaboxer),968(vboxsf),979(bluetooth),999(lpadmin)

gopika@kali:~$ who
gopika  seat0      2026-09-01 09:43 (:0)

gopika@kali:~$ w
09:46:14 up 3 min, 1 user, load average: 0.42, 0.60, 0.27
USER    TTY      FROM          LOGIN@   IDLE   JCPU   PCPU   WHAT
gopika  -        -             09:43    0.00s  0.04s  lightdm --session-child 13 24

gopika@kali:~$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root         1      0   0  09:42 ?        00:00:02 /sbin/init splash
root         2      0   0  09:42 ?        00:00:00 [kthreadd]
root         3      2   0  09:42 ?        00:00:00 [pool_workqueue_release]
root         4      2   0  09:42 ?        00:00:00 [kworker/R-rcu_gp]
root         5      2   0  09:42 ?        00:00:00 [kworker/R-sync_wq]
root         6      2   0  09:42 ?        00:00:00 [kworker/R-kvfree_rcu_reclaim]
root         7      2   0  09:42 ?        00:00:00 [kworker/R-slub_flushwq]
root         8      2   0  09:42 ?        00:00:00 [kworker/R-netns]
root         9      2   0  09:42 ?        00:00:00 [kworker/0:0-events]
root        10      2   0  09:42 ?        00:00:01 [kworker/0:0H-kblockd]
root        11      2   0  09:42 ?        00:00:00 [kworker/0:1-ata_sff]
root        12      2   0  09:42 ?        00:00:01 [kworker/u8:0-events_unbound]
root        13      2   0  09:42 ?        00:00:00 [kworker/R-mm_percpu_wq]
root        14      2   0  09:42 ?        00:00:00 [ksoftirqd/0]
root        15      2   0  09:42 ?        00:00:00 [rcu_preempt]
root        16      2   0  09:42 ?        00:00:00 [rcu_exp_par_gp_kthread_worker/0]
root        17      2   0  09:42 ?        00:00:00 [rcu_exp_gp_kthread_worker]
root        18      2   0  09:42 ?        00:00:00 [migration/0]
root        19      2   0  09:42 ?        00:00:00 [idle_inject/0]
root        20      2   0  09:42 ?        00:00:00 [cpuhp/0]
root        21      2   0  09:42 ?        00:00:00 [kworker/0:0-events]
root        22      2   0  09:42 ?        00:00:00 [kworker/1:0-events]
root        23      2   0  09:42 ?        00:00:00 [kworker/1:0H-kblockd]
root        24      2   0  09:42 ?        00:00:00 [kworker/u8:1-flush-8:0]
root        25      2   0  09:42 ?        00:00:00 [kworker/1:0-events]
root        26      2   0  09:42 ?        00:00:00 [kworker/1:0H-kblockd]
root        27      2   0  09:42 ?        00:00:00 [kworker/u8:1-flush-8:0]
```

5. Identify the current account's username, UID, GID, home directory and login shell from /etc/passwd.

```
grep "^$(whoami):" /etc/passwd
```

→ grep → Searches /etc/passwd for the current user's entry.

6. Determine how many account entries are in /etc/passwd.

```
wc -l /etc/passwd
```

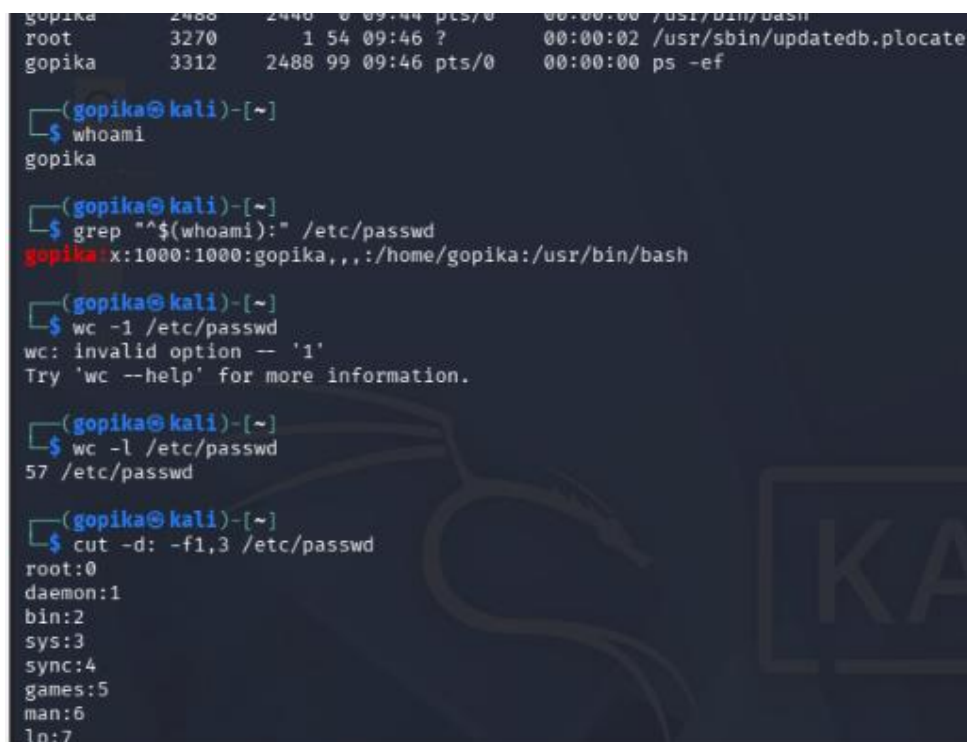
→ wc -l → Counts the number of lines/account entries.

7. Display only username and UID from /etc/passwd.

```
cut -d: -f1,3 /etc/passwd
```

→ cut → Uses : as the separator.

-f1,3 → Displays fields 1 (username) and 3 (UID).



```
gopika      2488      2448    0 09:44 pts/0    00:00:00 /usr/bin/bash
root        3270        1 54 09:46 ?        00:00:02 /usr/sbin/updatedb.plocate
gopika      3312      2488    99 09:46 pts/0    00:00:00 ps -ef

(gopika@kali)~$ whoami
gopika

(gopika@kali)~$ grep "^$(whoami):" /etc/passwd
gopika:x:1000:1000:gopika,,,:/home/gopika:/usr/bin/bash

(gopika@kali)~$ wc -l /etc/passwd
wc: invalid option -- 'l'
Try 'wc --help' for more information.

(gopika@kali)~$ wc -l /etc/passwd
57 /etc/passwd

(gopika@kali)~$ cut -d: -f1,3 /etc/passwd
root:0
daemon:1
bin:2
sys:3
sync:4
games:5
man:6
lp:7
```

8. Find all accounts whose UID is 0.

```
awk -F: '$3 == 0 {print $1}' /etc/passwd
```

→ awk -F: → Uses : as the field separator.

\$3 == 0 → Checks for UID 0.

print \$1 → Prints the username.

```
(gopika@kali)-[~]
$
(gopika@kali)-[~]
$ awk -F: ' $3 == 0 {print $1}' /etc/passwd
root
(gopika@kali)-[~]
$
(gopika@kali)-[~]
$ /etc/passwd
bash: /etc/passwd: Permission denied
(gopika@kali)-[~]
$ ls -l /etc/passwd
-rw-r--r-- 1 root root 3270 Aug 25 11:16 /etc/passwd
(gopika@kali)-[~]
$ sudo cat /etc/passwd
[sudo] password for gopika:
root:x:0:0:root:/root:/usr/bin/zsh
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
```

9. Compare /etc/passwd and /etc/shadow. Why are password hashes in /etc/shadow?

cat /etc/passwd OR sudo cat /etc/shadow

→ cat → Displays the file contents.

sudo → Runs the command with administrator privileges.

```
(gopika@kali)-[~]
$ sudo cat /etc/shadow
root:!:20675:0:99999:7:::
daemon:*:20675:0:99999:7:::
bin:*:20675:0:99999:7:::
sys:*:20675:0:99999:7:::
sync:*:20675:0:99999:7:::
games:*:20675:0:99999:7:::
man:*:20675:0:99999:7:::
lp:*:20675:0:99999:7:::
mail:*:20675:0:99999:7:::
news:*:20675:0:99999:7:::
uucp:*:20675:0:99999:7:::
proxy:*:20675:0:99999:7:::
```

```
pcscd:!:20675:0:99999:7:::
gopika:$y$j9T$KxLurVpbtppdg8XsHJ1k/1$iVtvjkCJQU9HEADDvNvtD8w2QcVjiZ4Ere4dA.X6mSD:20675:0:99999:7:::
(gopika@kali)-[~]
$ cat /etc/shadow
cat: /etc/shadow: Permission denied
(gopika@kali)-[~]
$ ls -l /etc/shadow
-rw-r----- 1 root shadow 1466 Aug 11 01:28 /etc/shadow
```

cat /etc/passwd

sudo cat /etc/shadow

ls -l /etc/passwd

ls -l /etc/shadow

The /etc/passwd file contains basic account information such as username, UID, GID, home directory, and login shell. It is normally readable by ordinary users. The password field usually contains x, indicating that the actual password hash is stored in /etc/shadow. /etc/shadow contains sensitive password hashes and password-aging information and is normally restricted to root/privileged access. Keeping hashes in /etc/shadow prevents ordinary users from easily obtaining them for offline password-cracking attempts.

Example/Expected result: Typical /etc/passwd entry: gopika:x:1000:1000:Gopika:/home/gopika:/bin/bash

Typical /etc/shadow entry: gopika:<password-hash>:<password-aging-fields>

Understanding the /etc/passwd fields

Field	Meaning	Example
1	Username	gopika
2	Password field / placeholder	x
3	UID	1000
4	GID	1000
5	Comment / user information	Gopika
6	Home directory	/home/gopika
7	Login shell	/bin/bash

Command	Expected output / result
whoami	gopika
id	uid=1000(gopika) gid=1000(gopika) groups=1000(gopika),27(sudo),24(cdrom)
who	gopika tty2 2026-09-01 09:10
w OR ps -ef	gopika tty2 09:10 0.00s bash OR UID PID PPID C STIME TTY TIME CMD
grep "^\$(whoami):" /etc/passwd	gopika:x:1000:1000:Gopika,,,:/home/gopika:/bin/bash
wc -l /etc/passwd	47 /etc/passwd
cut -d: -f1,3 /etc/passwd	root:0 daemon:1 bin:2 gopika:1000
awk -F: '\$3 == 0 {print \$1}' /etc/passwd	root
cat /etc/passwd OR sudo cat /etc/shadow	gopika:x:1000:1000:Gopika:/home/gopika:/bin/bash OR gopika:\$y\$j9T\$...

PART B — Creating and Verifying Users (20 Minutes)

1. Create alice, bob and charlie with a home directory for each.

```
sudo useradd -m alice
sudo useradd -m bob
sudo useradd -m charlie
```

→ sudo → Gives administrative permission.

useradd → Creates a user account.

-m → Creates the user's home directory.

2. Verify all three accounts were added to /etc/passwd.

```
grep -E '^(alice|bob|charlie):' /etc/passwd
```

→ grep -E → Uses an extended search pattern.

^(alice|bob|charlie): → Finds the three user entries.

3. Determine the UID assigned to each user.

```
id -u alice  
id -u bob  
id -u charlie
```

→ id -u → Displays only the UID of the specified user.

4. Determine Alice's home directory and default shell.

```
grep '^alice:' /etc/passwd
```

→ grep → Finds Alice's entry in /etc/passwd.

The entry contains her UID, GID, home directory and shell.

```
(gopika@kali)~[/linux/lab]  
$ sudo useradd -m alice  
[sudo] password for gopika:  
(gopika@kali)~[/linux/lab]  
$ sudo useradd -m bob  
(gopika@kali)~[/linux/lab]  
$ sudo useradd -m charlie  
(gopika@kali)~[/linux/lab]  
$  
(gopika@kali)~[/linux/lab]  
$ grep -E '^(alice|bob|charlie):' /etc/passwd  
alice:x:1001:1001::/home/alice:/bin/sh  
charlie:x:1003:1003::/home/charlie:/bin/sh  
(gopika@kali)~[/linux/lab]  
$  
(gopika@kali)~[/linux/lab]  
$ id alice  
uid=1001(alice) gid=1001(alice) groups=1001(alice)  
(gopika@kali)~[/linux/lab]  
$ id -u bob  
id: 'bob': no such user  
(gopika@kali)~[/linux/lab]  
$ id -u bob  
1002  
(gopika@kali)~[/linux/lab]  
$ id charlie  
uid=1003(charlie) gid=1003(charlie) groups=1003(charlie)  
(gopika@kali)~[/linux/lab]  
$  
(gopika@kali)~[/linux/lab]  
$ grep '^alice:' /etc/passwd  
alice:x:1001:1001::/home/alice:/bin/sh  
(gopika@kali)~[/linux/lab]  
$
```

5. Inspect /home and verify the required directories.

```
ls -ld /home/alice /home/bob /home/charlie
```

→ ls -l → Shows detailed directory information.

-d → Shows the directories themselves.

6. Set passwords for all three accounts.

```
sudo passwd alice  
sudo passwd bob  
sudo passwd charlie
```

- sudo → Gives permission to change the accounts' passwords.
- passwd → Sets or changes the specified user's password.

```

(gopika@kali)~[/linux/lab]
$ ls -l /home
total 16
drwx----- 5 alice  alice  4096 Sep  1 18:23  alice
drwx----- 5 bab   bab   4096 Sep  1 18:23  bab
drwx----- 5 charlie charlie 4096 Sep  1 18:23  charlie
drwx----- 23 gopika gopika 4096 Sep  1 18:16  gopika

(gopika@kali)~[/linux/lab]
$ ls -ld /home/alice
drwx----- 5 alice alice 4096 Sep  1 18:23 /home/alice

(gopika@kali)~[/linux/lab]
$ sudo passwd alice
New password:
Retype new password:
passwd: password updated successfully

(gopika@kali)~[/linux/lab]
$ sudo passwd bab
New password:
Retype new password:
passwd: password updated successfully

```

7. Switch to Alice using a login shell, verify, then return.

```

su - alice
whoami OR pwd OR id
exit → whoami

```

- su - alice → Starts Alice's login shell.
- whoami → Shows the current username.
- pwd → Shows the current working directory.
- id → Shows UID, GID and groups.
- exit → Leaves Alice's shell.
- whoami → Confirms the original account.

```

(gopika@kali)~[/linux/lab]
$ whoami
gopika

(gopika@kali)~[/linux/lab]
$ su - alice
Password:
$ whoami
alice
$ pwd
/home/alice
$ id
uid=1001(alice) gid=1001(alice) groups=1001(alice)
$ exit

(gopika@kali)~[/linux/lab]
$ whoami
gopika

```

Alice's account details

Information	Value
Username	alice
UID	1001
GID	1001

Home directory	/home/alice
Default shell	/bin/bash

Command	Expected output / result
sudo useradd -m alice sudo useradd -m bob sudo useradd -m charlie	No output on success.
grep -E '^(alice bob charlie):' /etc/passwd	alice:x:1001:1001::/home/alice:/bin/bash bob:x:1002:1002::/home/bob:/bin/bash charlie:x:1003:1003::/home/charlie:/bin/bash
id -u alice id -u bob id -u charlie	1001 1002 1003
grep 'alice:' /etc/passwd	alice:x:1001:1001::/home/alice:/bin/bash
ls -ld /home/alice /home/bob /home/charlie	drwx----- ... alice ... /home/alice drwx----- ... bob ... /home/bob drwx----- ... charlie ... /home/charlie
sudo passwd alice sudo passwd bob sudo passwd charlie	passwd: password updated successfully
su - alice whoami OR pwd OR id exit → whoami	alice OR /home/alice OR uid=1001(alice) ... then: gopika

PART C — Group Management and Access Control *(30 Minutes)*

1. Create the groups security and developers.

```
sudo groupadd security
sudo groupadd developers
```

- sudo → Gives administrative permission.
- groupadd → Creates a new group.

2. Add Alice and Charlie to security; add Bob to developers.

```
sudo usermod -aG security alice
sudo usermod -aG security charlie
sudo usermod -aG developers bob
```

- sudo → Gives administrative permission.
- usermod → Modifies an existing user account.
- aG → Appends the user to a supplementary group without removing existing groups.

```

(gopika@kali)-[~]
$ sudo groupadd security
groupadd: group 'security' already exists

(gopika@kali)-[~]
$ sudo groupadd developers

(gopika@kali)-[~]
$ getent group security
security:x:1006:

(gopika@kali)-[~]
$ getent group developers
developers:x:1007:

(gopika@kali)-[~]
$ sudo usermod -aG security alice

(gopika@kali)-[~]
$ sudo usermod -aG security charlie

(gopika@kali)-[~]
$

(gopika@kali)-[~]
$ sudo usermod -aG developers bob
usermod: user 'bob' does not exist

(gopika@kali)-[~]
$ sudo usermod -aG developers bab

```

3. Without switching accounts, verify group membership of all three.

```
id alice
id bob
id charlie
```

→ id → Shows each user's UID, GID and group memberships.

4. Display security and developers entries from /etc/group.

```
grep -E '^(security|developers):' /etc/group
```

→ grep -E → Uses an extended search pattern.

Finds only the security and developers entries.

5. Add Bob to security without removing him from developers.

```
sudo usermod -aG security bob
```

→ sudo → Gives administrative permission.

usermod -aG → Adds Bob to security while preserving developers.

6. Verify Bob now belongs to both groups.

```
id bob OR grep -E '^(security|developers):' /etc/group
```

→ id bob → Shows Bob's groups directly.

grep -E → Shows the security and developers entries from /etc/group.


```

(gopika@kali)-[~]
$ id alice
uid=1001(alice) gid=1001(alice) groups=1001(alice),1006(security)

(gopika@kali)-[~]
$ id bab
uid=1002(bab) gid=1002(bab) groups=1002(bab),1007(developers)

(gopika@kali)-[~]
$ id charlie
uid=1003(charlie) gid=1003(charlie) groups=1003(charlie),1006(security)

(gopika@kali)-[~]
$
(gopika@kali)-[~]
$ grep -E '^(security|developers):' /etc/group
security:x:1006:alice,charlie
developers:x:1007:bab

(gopika@kali)-[~]
$
(gopika@kali)-[~]
$ sudo usermod -aG security bab

(gopika@kali)-[~]
$
(gopika@kali)-[~]
$ id bab
uid=1002(bab) gid=1002(bab) groups=1002(bab),1006(security),1007(developers)

(gopika@kali)-[~]
$ grep -E '^(security|developers):' /etc/group
security:x:1006:alice,charlie,bab
developers:x:1007:bab

```

Command	Expected output / result
sudo groupadd security sudo groupadd developers	No output on success.
sudo usermod -aG security alice sudo usermod -aG security charlie sudo usermod -aG developers bob	No output on success.
id alice id bob id charlie	uid=1001(alice) ... groups=1001(alice),1004(security) uid=1002(bob) ... groups=1002(bob),1005(developers) uid=1003(charlie) ... groups=1003(charlie),1004(security)
grep -E '^(security developers):' /etc/group	security:x:1004:alice,charlie developers:x:1005:bob
sudo usermod -aG security bob	No output on success.
id bob OR grep -E '^(security developers):' /etc/group	uid=1002(bob) ... groups=1002(bob),1004(security),1005(developers) OR security:x:1004:alice,charlie,bob developers:x:1005:bob